International Journal of Extreme Manufacturing

3D-Printed Customizable Magnetic Soft Electronics Enabled by Nanometal Oxide-Assisted Laser Activation and Magnetic Liquid Metal

To cite this article before publication: Siying Li *et al* 2026 *Int. J. Extrem. Manuf.* in press https://doi.org/10.1088/2631-7990/aea667

3D-Printed Customizable Magnetic Soft Electronics Enabled by Nanometal Oxide-Assisted Laser Activation and Magnetic Liquid Metal

Journal:	*International Journal of Extreme Manufacturing*
Manuscript ID	IJEM-113379.R2
Manuscript Type:	Paper
Keywords:	3D-printed devices, laser, nano zinc oxide, magnetic liquid metal, antibacterial, magnetic or light drive

3D-Printed Customizable Magnetic Soft Electronics Enabled by Nanometal Oxide-Assisted Laser Activation and Magnetic Liquid Metal

Siying Li[1,§], Pengan Luo[1,§], Hao Lu[1], Haoran Xu[1], Feifan Yu[1], Jin Feng[1], and Tao Zhou[1,*]

[1] State Key Laboratory of Advanced Polymer Materials, Polymer Research Institute, Sichuan University, Chengdu 610065, China

E-mail: zhoutaopoly@scu.edu.cn

§ These authors contributed equally to this work.

Abstract

This study introduces a novel strategy based on laser-induced selective metallization technology (LISM) and magnetic liquid metal slurry infiltration to customise magnetic patterning on 3D-printed device surfaces. Incorporating nano zinc oxide (ZnO) into 3D printing resin endows the printed devices with laser activation capability and antibacterial properties. The patterned copper layer is obtained by LISM on the resin surface, and the magnetic slurry is formed by dispersing iron (Fe) particles in liquid metal, which can selectively infiltrate the copper layer, thereby imparting robust magnetic properties to the patterned conductive traces. The 3D-printed electronic devices based on LM-20%Fe@resin have excellent magnetic-thermal and light-thermal conversion capabilities. Demonstrated applications include torpedoes that can move flexibly in water through magnetic or light drives, and caterpillars that move directionally on planes and slopes under magnetic fields. Furthermore, the LM-20%Fe@resin possesses significant antibacterial properties, effective EMI shielding, low cytotoxicity, and enhanced sensing sensitivity due to the Fe particles. This enabled the fabrication of functional devices such as a wearable strain-sensing finger cuff for Morse code decryption and a piezoresistive pressure sensor with a 3D porous structure. Therefore, this method has a promising application in the field of soft robotics, wearable devices, and so on.

Keywords: 3D-printed devices, laser, nano zinc oxide, magnetic liquid metal, antibacterial, magnetic or light drive

Highlights

1. A strategy for achieving precise patterning using laser on various 3D-printed device surfaces was proposed.
2. The magnetic liquid metal slurry imparts deformable magnetism and conductivity to the resin.
3. The obtained circuit has a unique layered structure; liquid metal reacts with the Cu layer to form $CuGa_2$.
4. The obtained devices are non-toxic and possess good antimicrobial and EMI shielding capabilities, as well as dual magnetic–thermal and light–thermal responsiveness.
5. Magnetic particles restrict the flow of liquid metal, enhancing the sensitivity of the sensors.

1. Introduction

With the rapid advancement of soft robotics and smart systems, 3D-printed soft electronic devices with multi-functional responses have become a research hotspot [1-3]. Compared with conventional rigid electronics, soft devices exhibit excellent deformability and adaptability to complex environments [4, 5]. Soft 3D electronic devices with functional responses can convert external stimuli, such as light, electricity, and magnetic fields, into mechanical deformation, thereby completing the driving process [6-8].

Magnetic-driven and light-driven soft 3D devices do not need to be equipped with a power device, and the movement of these devices can be realized only by an external magnetic field or a light source [9, 10]. By integrating magnetic materials with 3D printing, it enables the fabrication of flexible devices with complex structures and controllable responses [11]. Various strategies have been reported to fabricate magnetic 3D-printed devices, including direct printing of magnetic composites, embedding prefabricated magnets, magnetic field-assisted printing, and post-processing magnetization [12-15]. However, these methods often face limitations such as insufficient patterning resolution, restricted design freedom, poor stretchability, and complex fabrication processes [16-18]. Therefore, fabricating high-resolution, complex magnetic patterns on 3D-printed soft devices remains a significant challenge.

Laser-induced selective metallization (LISM) presents a promising alternative, enabling the high-precision deposition of conductive patterns on non-metallic substrates (e.g., polymers, ceramics, and glass) without masks [19-21]. It is widely used in flexible electronics, microelectronic packaging, 3D circuit fabrication, and other fields [22]. Using laser manufacturing technology, specific regions on the 3D-printed device surface containing laser sensitizers are selectively scanned. The laser's light-thermal effect triggers the chemical reaction of the sensitizers within these regions, forming active catalytic seeds for electroless plating. These seeds subsequently initiate metal deposition (e.g., Au, Ag, Cu, and Ni) on the activated regions. The magnetic metal Ni layer can provide a certain magnetic property for 3D-printed devices, but its rigid nature makes it non-stretchable. When the flexible substrate is forced to deform, the substrate rebounds, causing the rigid metal layer to break and delaminate, reducing its conductivity and magnetism [23].

Dispersing soft or hard magnetic particles in a fluid carrier can form magnetic active slurries with deformability and magnetorheological effects [24, 25]. Liquid metal, with good biocompatibility, self-fluidity, and deformability, is an ideal choice for magnetic material carriers. Moreover, a 0.7–3 nm thick oxide layer spontaneously forms on the liquid metal surface in air, preventing further reactions between the liquid metal and air and maintaining the stability of the internal structure [26]. Such magnetic liquid metal active slurries not only retain the inherent advantages of liquid metal but also enable the liquid metal to act as a deformable host for magnetic particles [27]. Previous studies have demonstrated that liquid metal can be functionalized with magnetic particles to form magnetorheological slurries [28-30]. Furthermore, the addition of metallic particles to the liquid metal can significantly increase its light absorption capacity and light-thermal conversion efficiency, enabling applications in light-thermal drive [31-33]. Representative studies have shown that metal-doped liquid metals can generate efficient light-induced heating under near-infrared irradiation, highlighting their potential as multifunctional responsive materials [34-36]. Meanwhile, it has also been reported that adding fillers to liquid metal can increase the sensitivity for sensor applications. Adding fillers increases the viscosity of the material and restricts the flow of the liquid metal [37]. The interface formed between the filler and the liquid metal undergoes contact separation during stretching, altering pathways in the conductive network and leading to a significant change in resistance, thus increasing the sensitivity of the sensor [38, 39]. These findings indicate that incorporating magnetic particles into liquid metal can simultaneously impart the composite with magnetic responsiveness, photothermal activity, and enhanced sensing capability. Some 3D magnetic devices are fabricated by mixing magnetic particles into a polymer matrix (e.g., silicone rubber) and then casting the mixture into moulds [40, 41]. However, the magnetic and photothermal responses of such devices typically exhibit a uniform distribution throughout the entire structure, making it impossible to achieve magnetic-patterning customisation in desired regions. This severely limits their actuation performance and localized thermal control capabilities. In our strategy, magnetic liquid metal slurries can be patterned into any desired area on 3D surfaces, providing an ideal material platform for coupled magnetically and photothermally driven functionalities.

This work demonstrates a strategy for high-resolution magnetic patterning on 3D-printed device surfaces (**Figure 1**). The designed magnetic 3D-printed device is composed of three parts: resin (structural layer), Cu layer (interface layer), and Fe-doped liquid metal (functional layer). Among them, the 3D printing resin contains nano-ZnO, endowing the printed devices with laser activation capability and antibacterial properties. The liquid metal is tightly adhered to the Cu layer obtained by LISM through the alloying reaction, and the liquid metal is used as the carrier of magnetic materials. This process successfully attaches Fe particles to the 3D-printed device surface, achieving magnetic pattern customization. A comparison with existing fabrication methods (**Table S1**) reveals that this strategy simultaneously achieves high pattern resolution, high magnetic loading, strong mechanical flexibility, and high process efficiency, all of which are difficult to achieve simultaneously with conventional techniques. Moreover, it proposes a practical fabrication route aimed at addressing the challenges of achieving high-resolution magnetic patterning and multifunctional integration on complex 3D-printed device surfaces. By introducing a LISM-derived Cu interfacial layer and selective infiltration of a magnetic liquid metal slurry, this strategy can achieve high-resolution and multifunctional localized magnetic patterning on customizable 3D-printed device surfaces while also fully maintaining the structural deformability of the resin. Through the magnet's attraction to Fe particles, it is possible to control the liquid metal to be driven in the magnet's movement direction, thereby repairing circuits where mechanical damage has occurred. The work has successfully fabricated personalized soft electronic devices, including a small torpedo that can be magnetically and light thermally driven in the water, a caterpillar that can simulate climbing on land or slopes, and some flexible wearable sensors. In addition, the magnetic resin has excellent antibacterial properties and electromagnetic interference (EMI) shielding capabilities and is not toxic to human epidermal cells, which makes it advantageous for applications in the field of flexible wearable devices. This strategy, based on 3D printing combined with LISM technology and magnetic liquid metal slurry infiltration, offers a powerful and promising approach for fabricating multifunctional magnetic 3D-printed electronic devices with multiple materials, multiple sizes, and complex geometries.

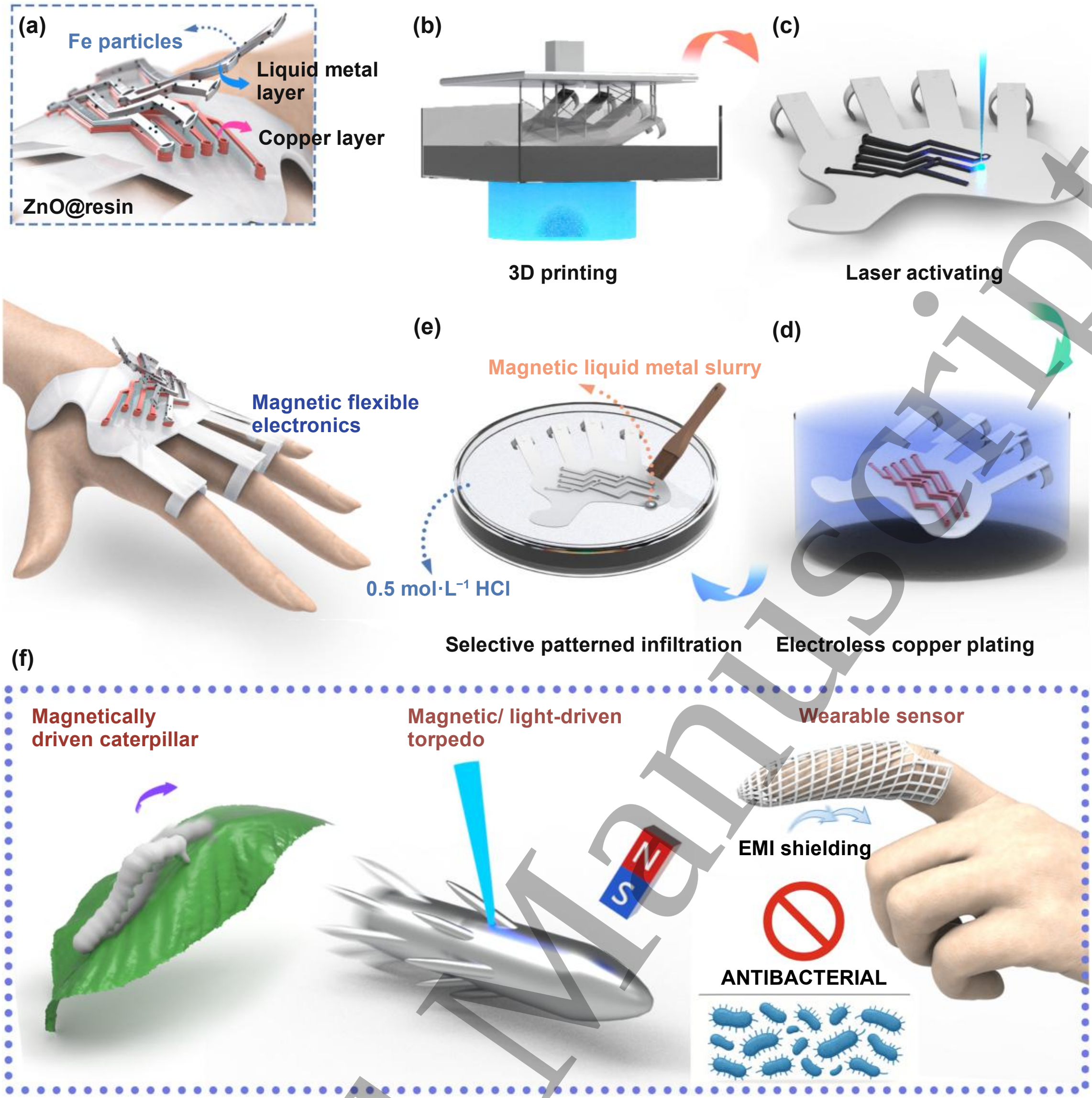

Figure 1. Schematic illustration of the fabrication and multifunctional applications of 3D-printed magnetic soft electronic devices. **(a)** The structure of the multifunctional 3D-printed electronic devices with magnetic patterns. **(b)** 3D-printed ZnO@resin devices obtained by SLA 3D printing. **(c)** 3D-printed ZnO@resin devices after UV laser activation. **(d)** Cu@resin devices prepared by the electroless copper plating process. **(e)** LM-Fe@resin devices prepared by brushing a magnetic liquid metal slurry in a 0.5 mol·L^{-1} HCl solution. **(f)** Representative application demonstrations of the 3D-printed magnetic soft electronic devices.

2. Results and discussion

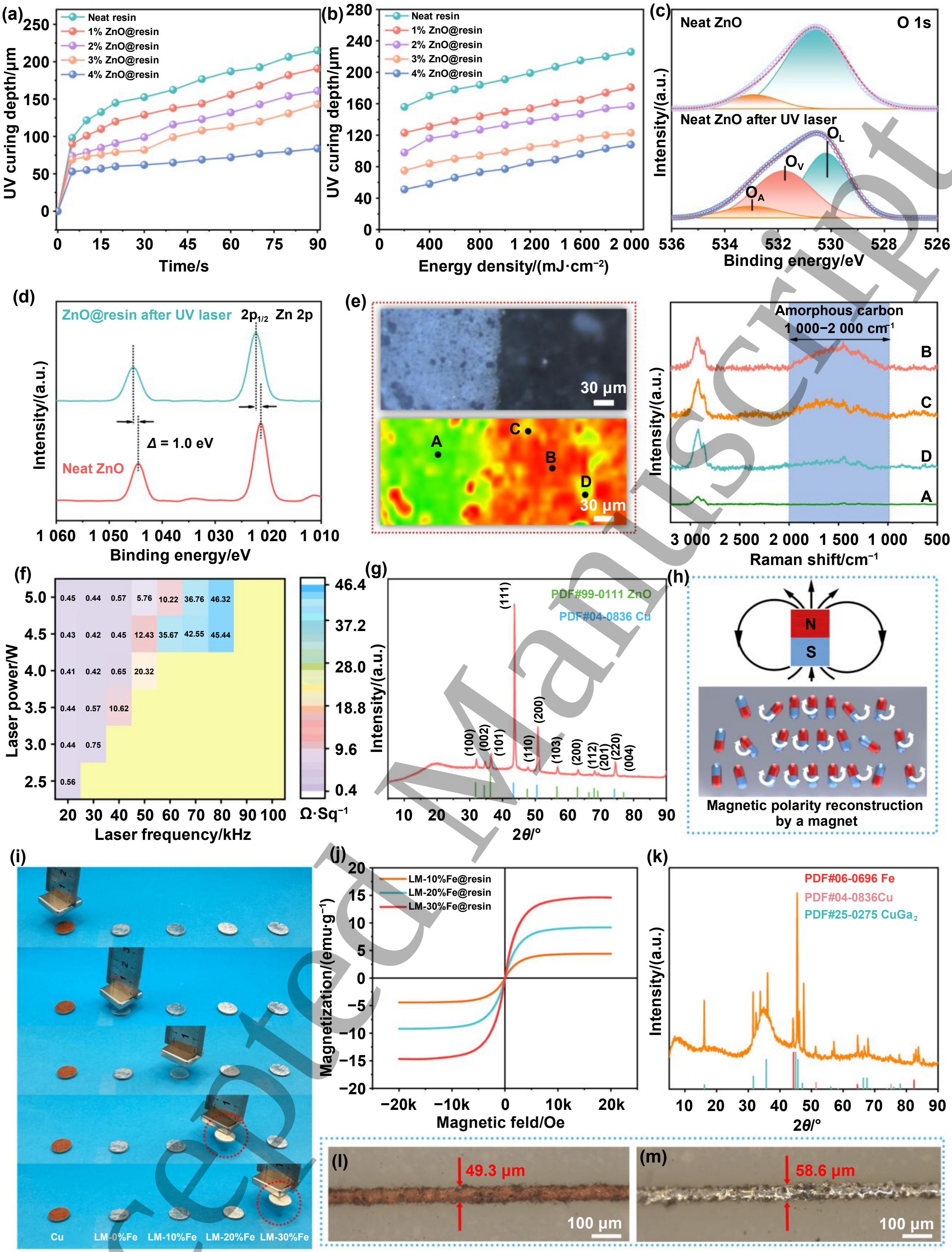

Figure 2. UV curing behavior, laser activation mechanism, and magnetic pattern formation during the fabrication of LM-Fe@resin. **(a)** The curing depth of the neat resin and ZnO@resin with different ZnO additions irradiated by a hand-held UV lamp (power: 3 W; irradiation distance: 5 cm; energy: 1 750 mJ·cm^{-2}). **(b)** The curing depth of the neat resin and ZnO@resin with different ZnO additions at different UV light energy densities. **(c)** O 1s XPS spectra of the neat ZnO powder before and after UV laser activation. **(d)** Zn

2p XPS spectra of the neat ZnO powder without UV laser activation and ZnO@resin after UV laser activation. **(e)** Micro-Raman imaging and representative micro-Raman spectra of the 4%ZnO@resin boundary after UV laser activation. **(f)** Square resistance heatmaps of the small squares in the Cu@resin matrix plot with different laser parameters. **(g)** XRD spectrum of Cu@resin. **(h)** Schematic diagram of the locally reconfigured magnetic polarity by a magnet. **(i)** Photographs of Cu@resin and LM-Fe@resin with different Fe additions attracted by a magnet. **(j)** Magnetic hysteresis curves of LM-Fe@resin with different Fe additions. **(k)** XRD spectrum of LM-20%Fe@resin. The narrowest line widths of Cu@resin **(l)** and LM-20%Fe@resin **(m)**.

First, we investigated the addition of the laser sensitizer nano ZnO in photosensitive resins. ZnO is a highly effective and safe spectral anti-ultraviolet agent that can scatter or reflect ultraviolet rays physically and then absorb and convert them into heat energy or fluorescence release. The morphology and particle size distribution of nano ZnO are shown in **Figure S1**, and the particle size of the ZnO used here is approximately (24.4 ± 5.7) nm. Excessive addition of ZnO possibly caused incomplete resin curing during the 3D printing process. Using |ow ZnO additions leads to the laser activation process to produce inadequate active seeds for ECP; thus, the complete Cu layer cannot be formed in the laser activation area during the subsequent ECP process. To explore the effect of ZnO on the light-curing properties of photosensitive resins, we mixed ZnO with neat resins at 1 wt%, 2 wt%, 3 wt%, and 4 wt% additions, and irradiated them with a hand-held UV lamp (power: 3 W; irradiation distance: 5 cm; energy: 1 750 $mJ \cdot cm^{-2}$). As shown in **Figure 2(**a**)**, the curing depth gradually decreased with the addition of more ZnO compared to the neat resin. Although 1%, 2%, and 3% ZnO@resin have a weaker effect on the curing depth, because of the insufficient addition amount, they cannot form a complete Cu layer in the subsequent ECP process with various laser parameters (**Figure S2**). The curing depth of the resin at different UV energy densities was further explored (**Figure 2(**b**)**), where the 4%ZnO@resin curing depth was still the lowest. At an energy density of 200 $mJ \cdot cm^{-2}$, the curing depth of the neat resin (156 μm) is about 3 times that of 4%ZnO@resin (51 μm). The SLA printer used in this work has a maximum laser energy density of 5 000 $mJ \cdot cm^{-2}$ so that printing can be performed normally at 4% ZnO addition [42]. Subsequently, by changing the laser parameters, a high-quality Cu layer can be obtained on the 4%ZnO@resin surface. Here, the following samples using 4%ZnO@resin are all called ZnO@resin.

With the action of a UV laser, generating oxygen vacancy (OV) in the laser sensitizer ZnO is essentially a photo-thermal-chemical process [43]. As shown in **Figure S3**, the local thermal and photochemical effects of the laser cause intense lattice vibration and the detachment of oxygen atoms from the lattice. Only ZnO with OV in the crystal lattice can trigger the formation of metallic Cu^0 particles during ECP [44]. To detect OV production, we carried out XPS detection of neat ZnO powder before and after UV laser activation. The O 1s

XPS spectra indicate (**Figure 2(c)**) that the peak of lattice oxygen (O_L) was detected at 530.5 eV for neat ZnO powder, and the peak corresponding to surface adsorbed oxygen (O_A) was detected at 533.0 eV. The defective oxygen peak (O_V) was detected at 531.8 eV in the ZnO powder after UV laser activation, demonstrating that the energy of the UV laser is high enough to break the Zn–O bond and cause oxygen atom detachment, resulting in the generation of OV in the ZnO lattice. However, since the oxygen vacancy is positively charged, the surrounding Zn^{2+} ions are bound by a stronger Coulomb bond because of the local charge imbalance, causing the binding energy shift to the high-energy direction [45]. Therefore, the XPS spectrum of Zn 2p shows (**Figure 2(d)**) that UV laser activation increases the binding energy of Zn $2p_{1/2}$ and Zn $2p_{3/2}$ by about 1.0 eV. **Figure S4** is the XRD pattern of neat resin and 4%ZnO@resin. Except for the amorphous broad peaks present in the range of 10°–30° in the neat resin itself, all the diffraction peaks match with the JCPDS No. 99-0111 of ZnO. The microscopic and Raman mapping images of the ZnO@resin boundary after laser activation are shown in **Figure 2(e)**. It shows representative microscopic Raman spectra obtained from the A, B, C, and D regions, where region A is the un-laser-activated region. Because the thermal effect of laser irradiation possibly caused local carbonization [46], we detected amorphous carbons with different degrees of carbonization within the broad peaks at 2 000–1 000 cm^{-1} in the B, C, and D regions. However, too high laser energy can lead to excessive ablation or graphitization, so exploring optimal laser parameters is also important.

At a fixed scanning speed of 500 $mm\cdot s^{-1}$, we changed the laser power and laser frequency, and constructed a matrix to explore the optimal laser parameters of ZnO@resin to obtain a high-quality Cu layer. As shown in **Figure S5**, the magnetic liquid metal slurry can only be coated on the high-quality Cu layer surface. The Cu layer in each small square of the matrix obtained by LISM has a different resistance, as illustrated in **Figure 2(f)**. When the laser parameters are 4.0 W and 20 kHz, the square resistance of the Cu layer is the lowest (0.41 $\Omega\cdot sq^{-1}$). The adjacent laser parameters of 4.0 W and 30 kHz, 4.5 W and 30 kHz also obtain a lower square resistance of 0.42 $\Omega\cdot sq^{-1}$. We tested the mechanical adhesion properties of the Cu layers obtained under optimal laser parameters (4.0 W and 20 kHz) according to the ASTM D3359 standard. As shown in **Figure S6,** the Cu layer showed no peeling in any of the tested regions, verifying that the Cu layer prepared under these conditions achieved the maximum 5B rating. The laser-activated ZnO@resin samples were stored for 0, 7, and 50 days before ECP. As shown in **Figure S7**, high-quality Cu layers can still be obtained even after 50 days of storage, indicating that the oxygen vacancies and their catalytic activity remain stable during long-term storage. We clearly detected Cu 2p peaks on the Cu layer surface (**Figure S8**). The XRD pattern also confirmed that not only the characteristic peaks of ZnO, but also the standard characteristic peaks of metallic Cu located at (111), (200), and (220) were detected after ECP, which matched with JCPDS No. 04-0836 (**Figure 2(**g**)**).

To investigate the optimal amount of Fe powder to be added to liquid metal, we configured different magnetic liquid metal slurries. 10 wt%, 20 wt%, and 30 wt% of Fe powder were added to the liquid metal and mixed well in 2 mol·L^{-1} HCl solution, respectively. Fe has no alloying reactions with liquid metal at room temperature, thereby greatly ensuring the stability of Fe magnetism in the magnetic liquid metal slurry. Afterward, the obtained magnetic slurries were coated on the disc-shaped Cu@resin surface, and then the discs were attracted by a NdFeB permanent magnet. Ferromagnetic materials have many tiny magnetic domains (spontaneous magnetization zones) inside, and when the magnet is brought close, the Fe powder is magnetized as a whole, resulting in an induced magnetic field [24]. Because the resin itself has a certain weight, the lower magnetic properties possibly do not make it attractable. As shown in **Figures S9 and 2(**i**)**, ZnO@resin, Cu@resin, and LM-0%Fe@resin without Fe powder have no magnetic properties; thus, they cannot be attracted to a magnet. In the case of resins infiltrated with a magnetic liquid metal slurry, LM-10%Fe@resin still cannot be attracted by a magnet due to its weak magnetic properties. Therefore, to obtain a high magnetic resin material, more than 20% of the Fe needs to be added to the liquid metal. **Figure 2(h)** illustrates that the magnetized Fe particles are distributed along the magnetic field lines because of the interaction between the poles (the N–S poles attract), and the magnetic field gradient force drives the Fe particles to move toward the magnet, and finally shows an apparent "attraction" phenomenon [24]. The Fe is a superparamagnetic material, and when a magnetic field is applied, magnetization forms an orientation arrangement, which disappears once the magnetic field is withdrawn [47]. The magnetic hysteresis curves (**Figure 2(j)**) reveal that as the content of Fe powder increases, the saturation magnetization of LM-Fe@resin significantly increases.

Measuring the adhesion of liquid metal by the rolling contact angle is an effective way to characterize its interaction with the contact surface. The difference between the forward angle (θ_a) and the backward angle (θ_r) is called the rolling angle (θ_m), which reflects the dynamic adhesion characteristics, where θ_m<10° represents a very low adhesion [48]. As shown in **Figure S10**, the θ_m of neat liquid metal LM-0%Fe and magnetic liquid metal slurry LM-20%Fe on the Cu layer are 24.4° and 46.1°, respectively, which indicates that both the liquid metal and magnetic liquid metal droplets have high adhesion on the Cu layer surface. **Figures 2(**l)–(m**) and**

S11 show digital photographs and 3D morphologies of the narrowest line widths of the Cu@resin and LM-20%Fe@resin circuits. The narrowest line widths of the Cu and LM-20%Fe circuits on the resin surface are 49.3 μm and 58.6 μm, respectively, indicating that the metal circuits prepared by UV LISM have excellent precision. To further evaluate the ultimate patterning resolution of the magnetic liquid metal patterning strategy, two parallel magnetic liquid metal circuits were fabricated, as shown in **Figure S12**. Stable, electrically isolated patterns can be achieved with a minimum line spacing of 40.23 μm. As shown in **Figure 3(**e**)**, we prepared a fish bone and a butterfly using 3D printing and selectively laser activated them in specific regions. The laser only scans and activates the area of the pattern we designed, and the color changes from white to grey. In the ECP process, a number of Cu particles are deposited in the laser-scanned area due to the participation of ZnO, and realize the selective patterning of magnetic liquid metal on the 3D device surface. As shown in **Video S1**, when the liquid metal slurry is coated on the 3D-printed butterfly resin surface, the magnetic slurry selectively and quickly adheres to the Cu region without adhering to the remaining un-activated regions. Therefore, the technology in this study is suitable for preparing high-precision magnetic liquid metal circuits or patterns on the surfaces of various 3D devices.

The XPS survey spectra of the elemental composition on each sample surface are shown in **Figures 3**(d) and **S13**, and we only detected the characteristic peaks of the liquid metal on the LM-20%Fe@resin surface. The high-resolution XPS spectra of Ga, In, and Sn are shown in **Figure 3**(a)–(c), where oxide layers can spontaneously form in air, leading to significant peaks for Ga_2O_3, In_2O_3, and SnO_2 in the XPS spectra. Since the detection depth of XPS is less than 10 nm, we cannot observe the signal peak of Fe because the Fe particles have been successfully encapsulated by liquid metal (**Figure S14**). We also performed XRD measurements on LM-20%Fe@resin (**Figure 2**(k)), and the spectrum shows broadened amorphous diffuse peaks at $2\theta \approx 30°$, corresponding to the disordered structure of the liquid metal [49]. The sharp peaks at $2\theta = 44.7°$ (110), 65.0° (200), and 82.3° (211) match perfectly with the α-Fe standard card (JCPDS No. 06-0696), indicating that the Fe powder still maintains the body-centered cubic (BCC) structure [50]. No peaks of the Fe–Ga compound were detected, indicating that these two compounds did not react. However, it is interesting to note the reaction of Cu with Ga in liquid metal to form copper-gallium intermetallic compounds ($CuGa_2$), which is accompanied by the Cu peak weakening [46]. This further demonstrates that the liquid metal, as the carrier of the magnetic material, successfully adheres the magnetic material to the resin surface through the alloying reaction of the liquid metal and copper [26, 51]. To analyze its elemental composition more directly, we performed EDS imaging of the boundary and cross-section of LM-20%Fe@resin. As shown in **Figures 3**(f) and (g), on the left is the un-laser activated region, the resin exhibits the original morphology, and the main elemental composition is C. After selective magnetic liquid metal coating, the right region has a high content of Ga, In, Sn, and Fe elements distribution. The energy-dispersive X-ray can penetrate the liquid metal oxide layers so that the metals Cu and Zn can still be detected at lower levels. The cross-sectional EDS image is identical (**Figure S15**(a)), with a magnetic liquid metal layer about 46.7 μm distributed mainly in the upper layer, and the lower layer is the resin substrate. The corresponding EDS curves and elemental contents are shown in **Figures S15**(b) and (c).

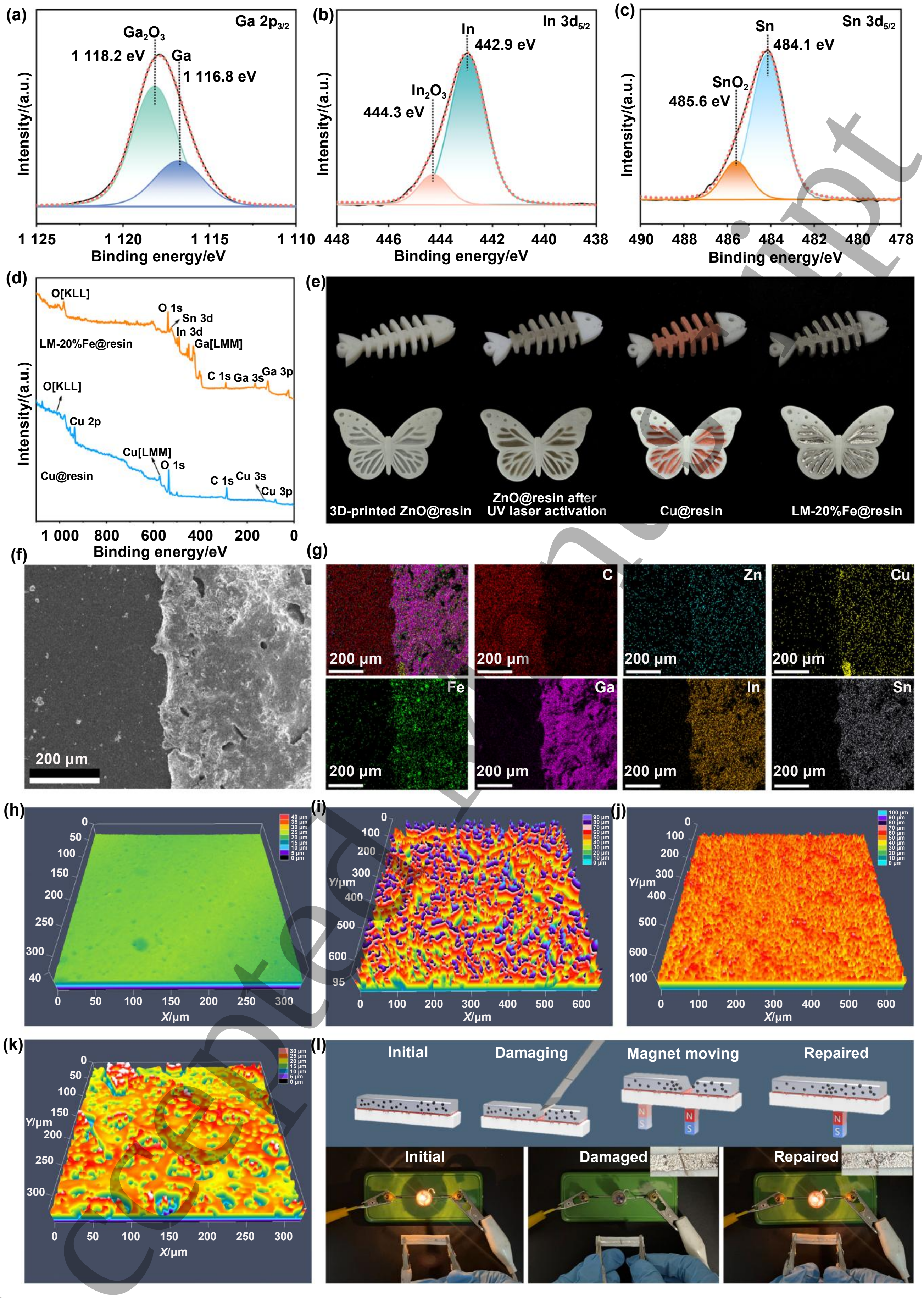

Figure 3. High-resolution XPS spectra of Ga **(a)**, In **(b)**, and Sn **(c)** on the LM-20%Fe@resin surface. **(d)** Surface XPS survey spectra of Cu@resin and LM-20%Fe@resin. **(e)** Photographs of a fish bone and a

butterfly prepared with ZnO@resin after UV laser activation, ECP, and magnetic liquid metal slurry infiltration. Surface boundary SEM image **(f)** and EDS image **(g)** of LM-20%Fe@resin. The LSCM images of the ZnO@resin before **(h)** and after laser activation **(i)**, Cu@resin **(j)**, and LM-20%Fe@resin **(k)**. **(l)** Schematic illustration and photographs of the healing process of an LM-20%Fe conductive circuit on the resin surface using a magnet.

Figure S16 shows the SEM images of ZnO@resin, Cu@resin, and LM-20%Fe@resin surfaces obtained at optimal laser parameters. The ZnO@resin surface after activation is distributed with holes of different sizes generated by laser ablation. These holes can be used as rivet locations during the ECP process, allowing the Cu^0 particles to be deposited in the holes and thus improving the adhesion between the Cu layer and the resin. After ECP, a continuous and dense metallic Cu layer is deposited on the resin surface. After being coated with the magnetic liquid metal slurry, the overall surface of the resin becomes flat, and some Cu^0 particles and square crystals of $CuGa_2$ can be observed through some holes. From its boundary SEM images (**Figure S17**), the magnetic liquid metal slurry can very well fill and cover some originally uneven places on the Cu layer. The magnetic liquid metal slurry can accurately cover the Cu layer with a clear boundary on the resin substrate, indicating that it has good selectivity for the Cu layer. The LSCM images and roughness curves of the different resin samples are shown in **Figures** 3(h)–(k) and **S18**, respectively. It can be observed that the ZnO@resin surface is very flat before laser activation, and its micro-roughness structure increases significantly after laser action, with pits of different sizes and uneven depths, and the depth of the pits can reach 100 μm. After ECP, more uniformly sized Cu particles were deposited on its surface and filled the laser-ablated holes, reducing its roughness. The LM-20%Fe@resin surface exhibits fluid-like properties, and although unevenness still exists, the pit depth is always below 30 μm, which is consistent with the results of the SEM image. The surface roughness of the resin directly affects its water contact angle, as shown in **Figure S19**. After laser activation, the surface exhibits super-hydrophobicity of about 140.1°, while the reduced surface roughness of Cu@resin and LM-20%Fe@resin results in lower contact angles of approximately 90.2° and 99.4°, respectively.

Figure 3(l) shows the schematic and digital photographs of using a magnet to repair a circuit that has been mechanically damaged. We prepared an LM-20%Fe conductive circuit on the resin surface, which had a high conductivity and could light a small light bulb successfully. After the circuit was cut using a knife, the small bulb went out, and we used a magnet to move it under the resin. Through the magnetic attraction of the magnet, the Fe particles aggregate and cause the fluid liquid metal carrier to move to the magnetic field direction, thus filling the mechanically damaged gap and allowing the small bulb to be lit again [9]. The top right corner of the digital photograph shows an enlarged view of the damaged and repaired circuits.

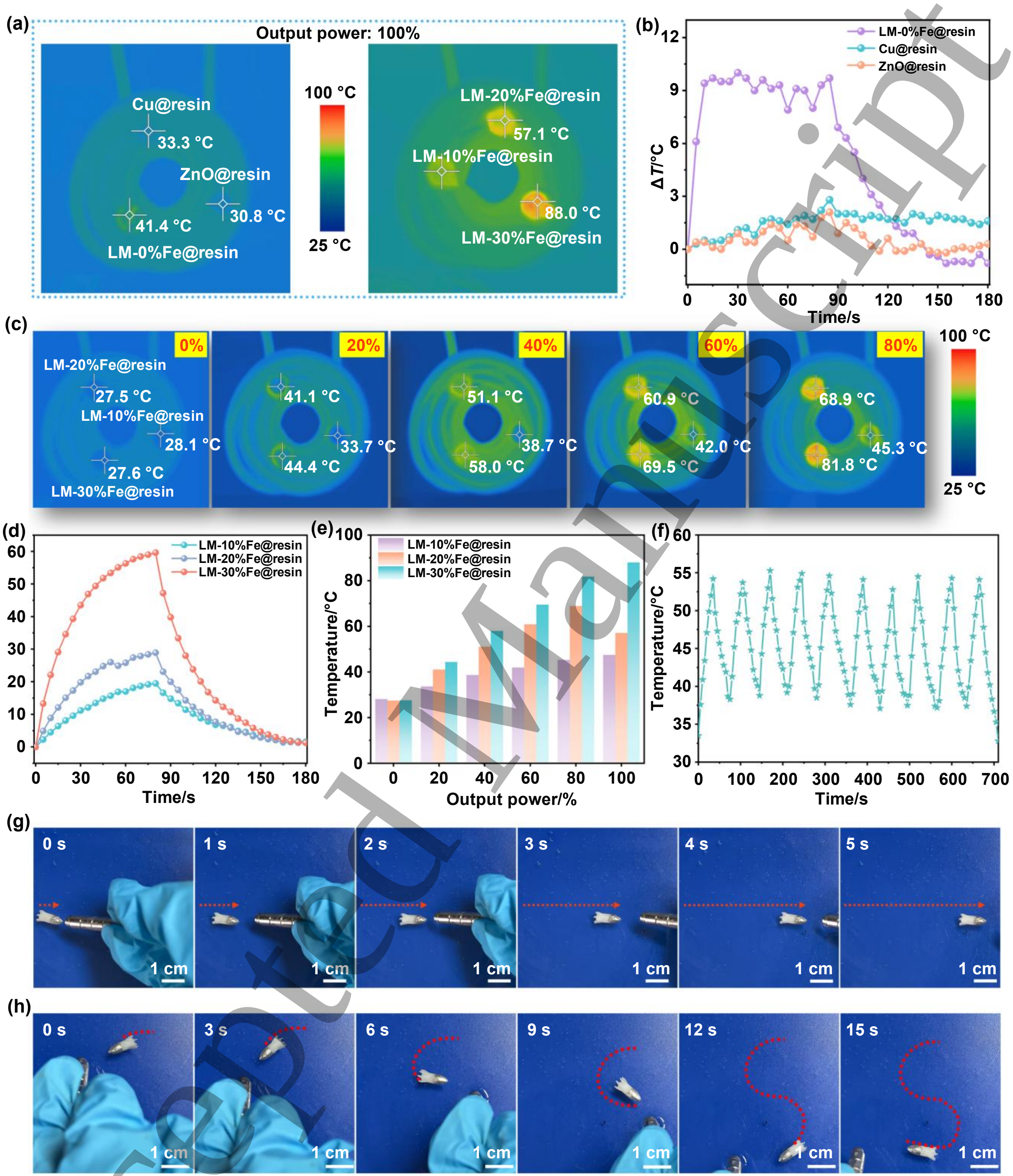

Figure 4. Magnetic–thermal conversion performance and magnetically driven movement of LM-Fe@resin. **(a)** Infrared thermal images of different resin samples with an alternating magnetic coil at 100% output power. **(b)** Changes in the temperature of the ZnO@resin, Cu@resin, and LM-0%Fe@resin tested for 3 min at 100% output power. **(c)** Infrared thermal images of LM-Fe@resin with different Fe additions under different output powers. **(d)** Changes in the temperature of LM-Fe@resin with different Fe additions tested for 3 min at 100% output power. **(e)** The highest temperatures for LM-Fe@resin with different Fe additions under different output

powers. **(f)** The cyclic heating/cooling curve of LM-20%Fe@resin at 100% output power. Photographs of a 3D-printed small torpedo in linear **(g)** and S-shaped **(h)** motion driven by a magnetic force.

Under the action of an alternating magnetic field, the hysteresis heating effect produced by the magnetized resin can convert electromagnetic energy into thermal energy [52], which is the key to determining the heat production capacity of the magnetized resin. We tested the magnetic–thermal conversion ability of different resin samples. Different resin samples were placed on a sample stage with an alternating magnetic coil, as shown in **Figure 4**(a). The magnetic field was applied to them for 80 s and then disconnected, and the temperature was recorded at 5 s intervals. At an output power of 100%, because of the lack of induced heat-generating particles in the ZnO@resin and Cu@resin samples, they have no thermal response in the magnetic field, and their surface temperatures have almost no change (**Figure 4**(b)). Although LM-0%Fe@resin has no magnetism, it produces eddy currents inside the liquid metal. The heat generation efficiency and poor thermal conductivity of the liquid metal are low in the magnetic field, resulting in its temperature rising by 9.4 °C in 10 s (the final surface temperature reaches 41.4 °C). For the LM-Fe@resin samples with the addition of Fe, under the same magnetic field conditions, the greater the concentration of magnetic material is, the faster it heats up, corresponding to a higher temperature, which is consistent with the results of the hysteresis loop test (**Figure 4**(d) and **Video S2**). The surface temperature of the LM-30%Fe@resin sample can reach 88.0 °C at 100% output power. The LM-Fe@resin samples were then heated with electromagnetic coils of different powers (0%, 20%, 40%, 60%, and 80%), as shown in **Figure 4**(c) and (e). The results show that the higher the output power of the electromagnetic coil is, the higher the magnetic field strength, causing higher temperatures corresponding to magnetic samples. Therefore, the temperature of the magnetic resin in the alternating magnetic field can be controlled by changing the Fe content and the output power. Finally, we tested the magnetic-thermal stability of LM-20%Fe@resin (output power: 100%), and it can be observed from **Figure 4**(f) that the magnetic-thermal heating performance of LM-20%Fe@resin is stable during repeated heating up and cooling down. We printed a small resin torpedo using 3D printing (**Figure S21**(a)), and magnetized the torpedo head so that it could be magnetically driven in the water. As shown in **Figure 4**(g)–(h) and **Video S3**, under the attraction of an external magnetic field, the small torpedo can be magnetically driven to perform a variety of motions in the water, including both linear and S-shaped motions.

Inspired by the fact that caterpillars in nature can move forward by wriggling, we fabricated a simple caterpillar robot using 3D printing. During the process of movement, the caterpillar's body alternates between two states: contraction and extension. The movement will be completed by an alternating repetition of these two states [53]. Since the soft resin caterpillar presents a curved state in its natural state, we locally magnetize the abdomen of the caterpillar. The abdomen stretches under the attraction of the magnetic field, which is similar to the state of the caterpillar when it crawls. As demonstrated in **Figure 5**(a), the caterpillar deforms under a stronger magnetic field, similar to the muscle contraction that causes the caterpillar to move forward. When the magnetic field is weakened, the caterpillar shrinks back to its initial state due to the elasticity of the resin itself. During the contraction process, the caterpillar's tail is larger, so there is more friction, and when the head remains still and the tail contracts slowly, it allows the caterpillar to move forward towards the head direction [53]. As shown in **Figure 5**(b) and (c) and **Video S4**, the displacement of the caterpillar in the horizontal direction along the X direction increased by approximately 3.5 cm in 15 s. In addition, we tested the caterpillar's climbing ability, as shown in **Figure 5**(d) and (e). Under the action of a magnetic field for 15 s, the caterpillar can climb upwards along a slope of about 30°, and the displacements of the caterpillar along the X and Z directions are approximately 2.5 cm and 1.0 cm, respectively. Experiments have confirmed that the alternating appearance of the magnetic field can realize the directional movement of the caterpillar on flat ground and the slope. In addition to the caterpillar structure, 3D printing technology can also be used to prepare a variety of small walking robots with bionic shapes, which are expected to perform some special tasks in a narrow space. Through local magnetization, these bionic robots can crawl on rough surfaces under a magnetic field and do not require an external power source.

Nano ZnO can not only be used as a laser sensitizer to induce ECP but also as a spectral antibacterial agent. It can catalyze the production of reactive oxygen species (ROS) to destroy bacterial cell membranes, proteins, and DNA. Meanwhile, ZnO can release Zn^{2+} slowly under wet conditions, interfering with bacterial enzyme systems and damaging membrane stability [54]. The smaller particle size and the larger specific surface area lead to higher release efficiencies of ROS and Zn^{2+}, thereby enhancing the antibacterial ability. We tested the antibacterial properties of LM-20%Fe@resin. As shown in **Figure 5**(f) and (g), LM-20%Fe@resin strongly inhibited the growth of the Gram-negative bacterium *E. coli* and the Gram-positive bacterium *S. aureus*, reaching more than 99% resistance to both. Control experiments revealed that ZnO@resin and Cu@resin exhibited moderate antibacterial activities (79%−85% against *E. coli* and 80%−81% against *S. aureus*), whereas LM-20%Fe@resin achieved significantly higher antibacterial efficiency. These results indicate that the superior antibacterial performance of LM-20%Fe@resin cannot be attributed to either the ZnO or the Cu layer alone, but rather arises from the synergistic effect of multiple components. Specifically, liquid metal's high surface tension and mobility can encapsulate bacteria and disrupt cell membrane integrity [55]. Fe particles, although exhibiting relatively weak intrinsic bactericidal activity, can promote bacterial membrane disruption through surface adsorption [56]. So, the antibacterial property of LM-20%Fe@resin is a synergistic effect between the metals. From the morphological changes in the bacteria (**Figure S20**), we can see that the untreated control group shows intact, full-length, long-rod-shaped *E. coli* and ball-shaped *S. aureus*. After

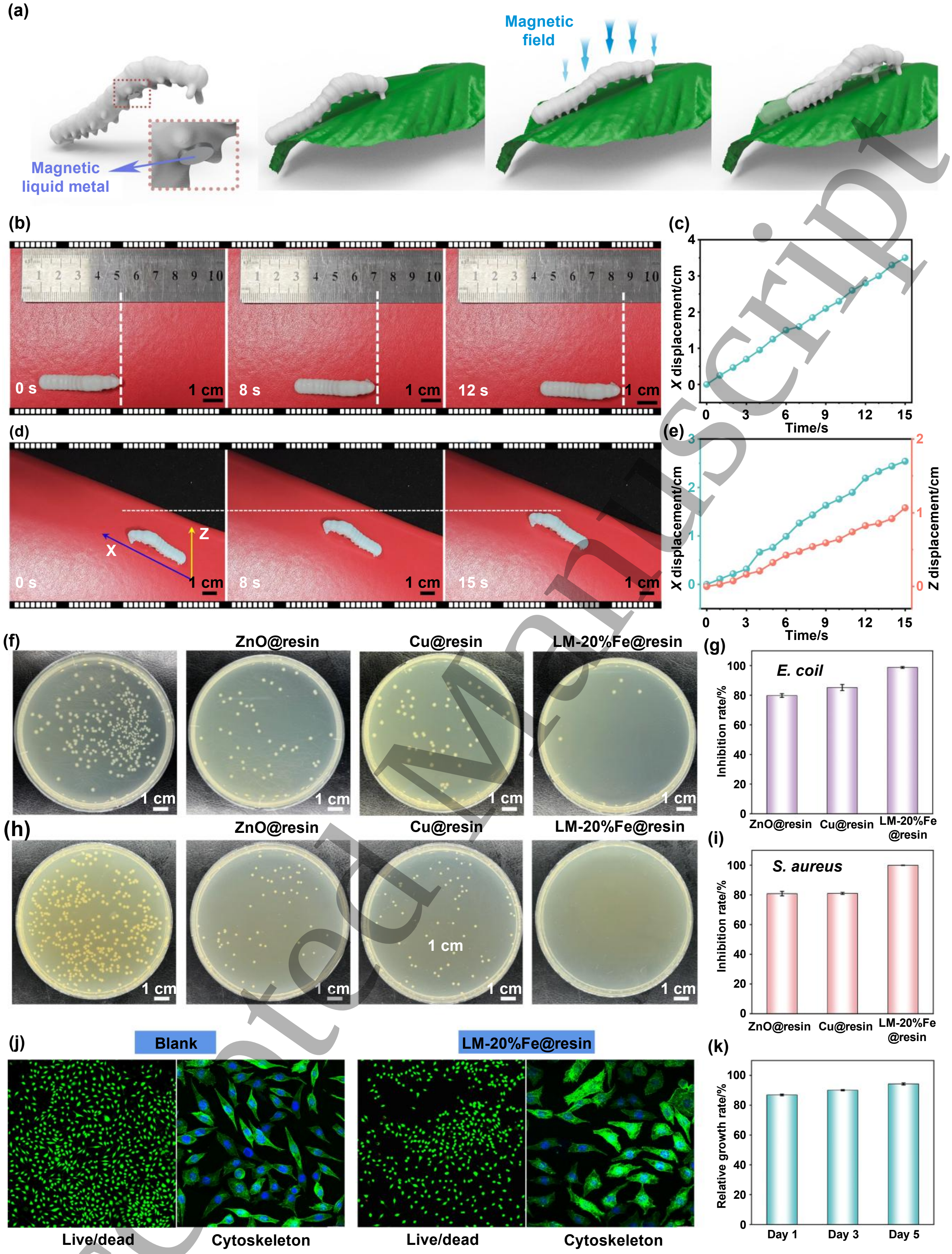

Figure 5. Magnetic actuation and biocompatibility of LM-20%Fe@resin and antibacterial comparison of different functional resin samples. **(a)** Schematic illustration of a 3D-printed caterpillar being driven by a magnetic force. Photographs of a 3D-printed caterpillar being driven on flat ground **(b)** and the climbing distance along the X-direction **(c)**. Photographs of a 3D-printed caterpillar being driven on a slope **(d)** and the

climbing distance along the X and Z directions **(e)**. Photographs showing the antibacterial activity of ZnO@resin, Cu@resin, and LM-20%Fe@resin against *E. coli* **(f)** and *S. aureus* **(h)** and the corresponding inhibition rates against *E. coli* **(g)** and *S. aureus* **(i)**. **(j)** The live/dead staining and cytoskeletal staining of L929 cells treated with LM-20%Fe@resin. **(k)** The cell viability of L929 cells treated with LM-20%Fe@resin.

LM-20%Fe@resin treatment, the bacterial morphology was severely changed, and the cell membranes appeared to shrink, wrinkle, or even rupture. The biosafety of LM-20%Fe@resin was assessed by L929 cell morphology analysis and viability. As shown in **Figure 5**(k), cell survival remained above 85% at all time points tested. The cells in the experimental group maintained a spindle-shaped morphology, with no changes in cell morphology or skeleton deformation, which was the same as the morphology of healthy fibroblasts in the control group (**Figure 5(j)**). These results indicate that LM-20%Fe@resin has good cytocompatibility, suggesting that they are non-toxic and can be used as a substrate for preparing wearable devices.

To verify the light-thermal properties of the magnetic resin, we magnetized the tail of a 3D-printed resin torpedo (**Figure S21**(b)), making the tail of the torpedo with light-thermal conversion and drive capability. First, we irradiated the resin surface with a near-infrared (NIR) laser and observed the temperature change. When LM-0%Fe@resin and LM-20%Fe@resin were exposed to the 808 nm NIR laser, their temperature changes increased with the increase of laser power. When LM-0%Fe@resin was irradiated at 1.0 W for 1 min, the resin temperature difference was only approximately 11 °C (**Figure 6**(a)). This is due to the fact that neat liquid metal is silver in color and has a smooth surface with high specular reflectivity, and this high reflectivity characteristic makes it commonly used in the area of smart reflective coatings. When the Fe particles in the LM-20%Fe@resin are irradiated by a laser, the free electrons on the LM-20%Fe@resin surface oscillate collectively (plasmon resonance), which can significantly absorb light energy and convert it into heat energy. Therefore, as shown in **Figure 6**(d), LM-20%Fe@resin has an excellent light-thermal conversion capability compared to LM-0%Fe@resin, and the lower power of the NIR laser still has an excellent thermal response capability. As shown in **Figure 6**(b) and (c), the light-thermal performance is related to the NIR laser power; the temperature difference on the LM-20%Fe@resin surface can reach approximately 38 °C (maximum temperature of 64.9 °C) at 1 W. Considering that local temperatures during photothermal drive can reach 64.9 °C, the thermal stability of the photosensitive resin substrate was further evaluated (**Figure S22**). Thermogravimetric analysis (TGA) results show that both the neat resin and ZnO@resin are thermally stable below 280 °C, with decomposition temperatures above 300 °C. As the photothermal temperature (64.9 °C) is far below this threshold, thermal degradation or structural instability cannot occur during operation. In addition, after 10 NIR laser on/off cycles, LM-20%Fe@resin exhibited significant light-thermal stability, with negligible temperature decay after cyclic irradiation (**Figure 6(**e**)**). To assess the influence of long-term thermal exposure, tensile specimens of the cured resin were subjected to thermal aging at 65 °C for 0, 3, and 7 days. The mechanical testing revealed no significant decrease in the overall tensile properties, indicating that long-term thermal cycling at this temperature does not degrade the mechanical integrity or flexibility of the resin (**Figure S23**). These results confirm that LM-20%Fe@resin has excellent light-thermal conversion capability, good thermal stability and repeatability, which provide a theoretical basis for further evaluating its light-thermal driving capability.

Light-thermal materials can convert light energy into heat energy when they are exposed to light, resulting in local temperature changes [52]. The small torpedo is magnetized only at the tail and irradiated only by a directional light source. The resulting temperature change is asymmetric, producing a temperature gradient and changing the surface tension of the liquid in the corresponding region. Liquid at lower surface tension would move along the tension gradient and change direction towards the higher surface tension, resulting in local liquid flow and driving the torpedo. The water contact angles of both the ZnO@resin and the LM-20%Fe@resin are >90°, indicating that all parts of the small torpedo are hydrophobic. The fine movement of the torpedo is controlled by controlling the local position of the light irradiation. When the torpedo's tail is irradiated with an 808 nm NIR laser at 1.0 W, the heat-flow trajectory is clearly recorded by infrared thermography, as shown in **Figure** 6(f)–(h) and **Video S5**. In addition to conducting linear forward motion by controlling light irradiation, the torpedo can be controlled to turn left/right by irradiating the left and right ends of the tail, respectively. Under light, the torpedo tail forms a high heat area to produce an asymmetric temperature gradient, and the surface tension difference is used to transfer the torpedo to the high surface tension region. Based on these results, the small torpedo prepared by LM-20%Fe@resin has an excellent dual response of light and magnetic, which can be magnetically driven by head magnetization and light-thermal driven by tail magnetization.

In addition, the thermal response of LM-20%Fe@resin in preparing wearable devices can also be applied to medical wearable thermotherapy devices in the future, such as local thermotherapy for tumors and joint thermotherapy to relieve pain and swelling [57]. The EMI shielding efficiency of different resins was also tested. **Figures 6(**i)–(k) and **S24** show the different resins' test results of the attenuation ability of incident electromagnetic waves in the X-band frequency range. ZnO@resin before or after laser activation has no EMI shielding efficiency. The average EMI shielding values of Cu@resin, LM-0%Fe@resin, and LM-20%Fe@resin are 40.5 dB, 74.7 dB, and 93.6 dB, respectively. The SE_R values are much larger than the corresponding SE_A values, which confirms that magnetic liquid metal resins rely mainly on the reflection mechanism to shield electromagnetic waves. The high shielding capacity of LM-20%Fe@resin is due to the unique dual-function synergistic mechanism of the magnetic liquid metal slurry, which combines the high electrical conductivity of the liquid metal with the magnetic loss characteristics of the magnetic component [58]. Liquid metal reflects part of the electromagnetic wave, and magnetic particles increase the magnetic permeability; thus, the EMI shielding ability of LM-20%Fe@resin is higher than that of LM-0%Fe@resin.

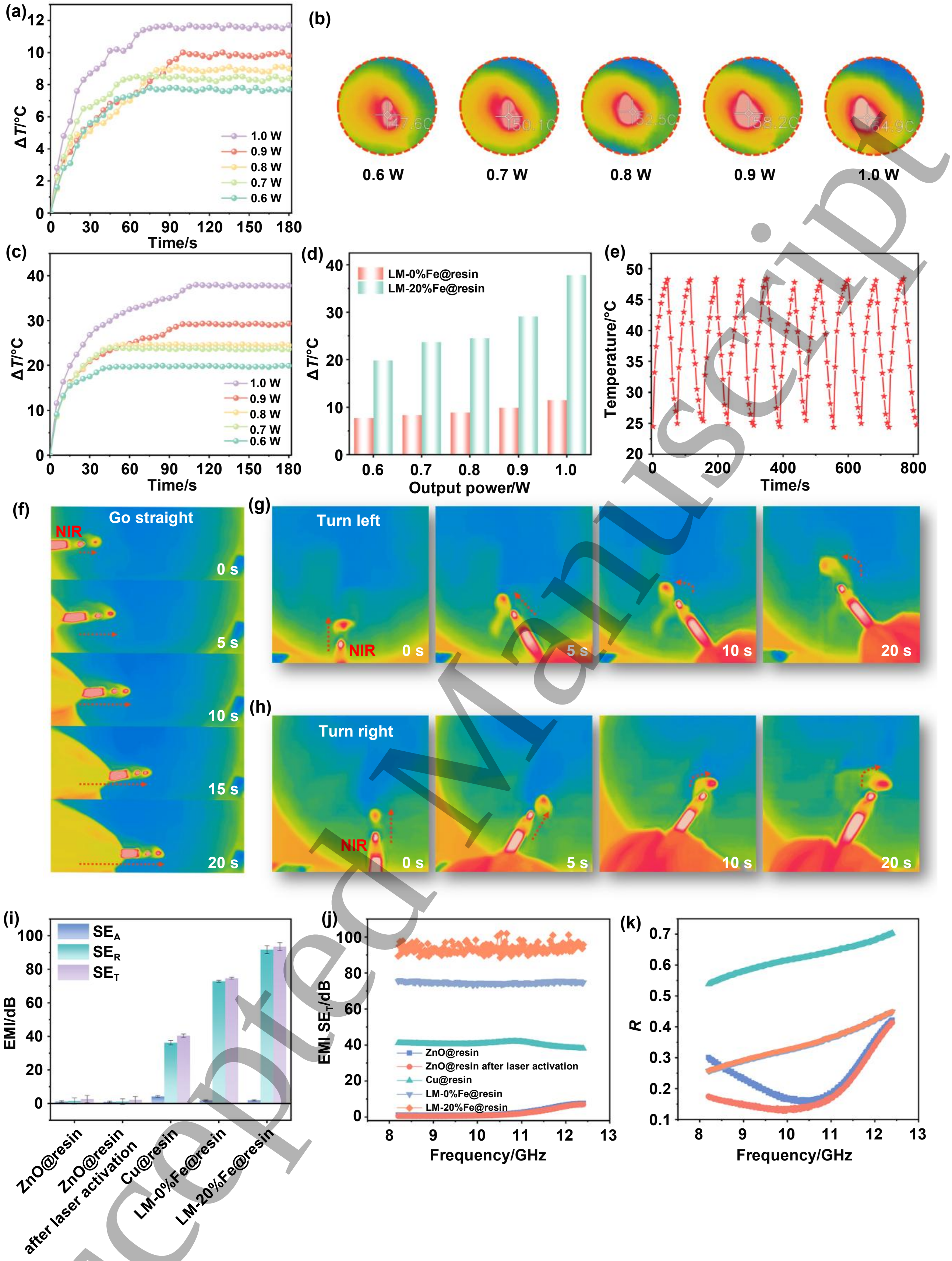

Figure 6. Light–thermal conversion and light-driven movement of LM-20%Fe@resin and EMI shielding performance comparison of different resin samples. **(a)** The temperature changes of LM-0%Fe@resin under 808 nm NIR laser irradiation at different powers. **(b)** Infrared thermal images of LM-20%Fe@resin under different NIR laser powers. **(c)** The temperature changes of LM-20%Fe@resin under an 808 nm NIR laser

irradiation at different powers. **(d)** Comparison of the temperature changes of LM-0%Fe@resin and LM-20%Fe@resin under different NIR laser powers. **(e)** The cyclic irradiation curve of LM-20%Fe@resin under a 0.6 W NIR laser. The small torpedo moved straight forward **(f)** and turned left **(g)** or right **(h)** under a 1.0 W NIR laser. **(i)** Comparison of SE_T, SE_R, and SE_A of different resin samples. EMI SE_T **(j)** and reflection loss (R) values **(k)** of different resin samples.

Liquid metal-based stretchable circuits/sensors are a revolutionary technology in the field of flexible electronics, with the advantages of simultaneously achieving high conductivity, large deformation tolerance, and dynamic adaptivity [51, 59]. As shown in **Figure 7**(a), the Cu@resin circuit can light up the small bulb in the initial state, but when the circuit is in a bending and twisting state, the brightness of the small bulb decreases significantly, indicating that its resistance increases or even breaks the circuit (**Figure 7**(c)). When stretching, the circuit is completely disconnected and the small bulb goes out, because the Cu layer produces numerous cracks when stretched, resulting in an increase in resistance (**Figure S25**(a)). However, the LM-20%Fe@resin circuit can still maintain the high brightness of the small bulb under the "initial," "bending," "twisting," and "stretching" states, indicating that its resistance changes slightly (**Figure 7**(b) and **Video S6**). The LM-20%Fe@resin circuit also has excellent conductivity over a wide range of strains, as shown in **Video S7**. Studies have shown that adding fillers to liquid metal increases the viscosity of the material, restricting the flow of liquid metal and thereby improving its sensitivity as a strain sensor [39]. Therefore, we measured the resistance changes of the LM-0%Fe@resin circuit and the LM-20%Fe@resin circuit under different tensile strains. As shown in **Figure 7**(d), the LM-0%Fe@resin circuits exhibit lower resistance change values of 8.1%, 15.5%, 34.5%, and 66.2% at 10%, 20%, 30%, and 40% stretching, respectively, because the mobility of the neat liquid metal can quickly fill the gaps during stretching (**Figure S25**(b)). The addition of Fe particles causes the resistance change value to increase under the same strain conditions (**Figure 7**(e)). This is due to the fact that the interface formed between the Fe particles and the liquid metal undergoes contact separation during stretching, altering the pathways in the conductive network (**Figure S25**(c)) and increasing the sensitivity of the strain sensor, ultimately leading to a larger change in the resistance of the LM-20%Fe@resin compared to the LM-0%Fe@resin. To investigate the long-term stability of the sensor, we conducted a 500-cycle resistance response test of the LM-20%Fe@resin circuit at 30% tensile strain (**Figure 7**(f)). The results show no significant change in the shape of the response curve or the change value, indicating the good durability of the LM-20%Fe@resin strain sensor. As shown in **Figure S26**, we stored the LM-20%Fe@resin circuit at room temperature for 6 months, with the resistance showing almost no change, indicating that the LM-20%Fe@resin circuit has good long-term conductivity stability. The electromechanical reliability of the

LM-20%Fe@resin circuit without encapsulation was systematically evaluated under mechanical vibration, perspiration-induced corrosion, varying pH, and temperature fluctuations (**Figure S27**). In all the cases, the LM-20%Fe@resin circuits maintain stable, low resistance, and only slight or negligible changes in resistance were observed.

To investigate whether this technique has a serious impact on the mechanical properties of the resin, we tested the mechanical properties of the ZnO@resin and LM-20%Fe@resin, respectively (**Figure S28**). The results show that the mechanical properties of the prepared LM-20%Fe@resin are only slightly reduced compared with ZnO@resin, and still meet the requirements for practical use. Because the LM-20%Fe@resin strain sensor exhibits high sensitivity, high stretchability, and excellent durability, it can be used to detect different activities of the human body. Based on this, we prepared a strain-sensing finger cuff by 3D printing for application in spy communication tools. By detecting the regular bending of the fingers, information is efficiently transmitted by relying on the Morse code principle (**Figure 7**(g)). **Figure 7**(h) shows a digital photograph of the strain-sensing finger cuff. The resistance increases when bending, and long-term bending can output a horizontal signal in the Morse code table, while short-term bending outputs a point signal in the Morse code table **(Figure 7**(i)). The Morse code of the English alphabet is composed of horizontal signals and point signals arranged in a regular manner, and the information can be transmitted by regularly outputting the Morse code. As shown in **Figure 7**(j)–(l), when the Morse code is emitted by bending the finger ("SOS", "RETREAT", and "HELP"), the strain-sensing finger cuff immediately produces a corresponding resistance change. The use of oscilloscopes and other instruments to record the pattern of resistance change in real time and then decode it according to the Morse code table to achieve information transfer. Afterward, we designed a piezoresistive pressure sensor with a three-dimensional porous skeleton structure through 3D printing, which can convert different stress changes into different resistance signals. As shown in **Figure 7(**m**)**, as the pressure applied to the pressure sensor device increases, the resistance value change increases gradually, the output signal is stable, and the response peak is constant.

Because we can only perform LISM on the material surface, when the three-dimensional porous skeleton structure is compressed, the original conductive path is destroyed because of the deformation of the porous structure, forcing the electron transport path to become longer, thereby increasing the resistance of the material [39]. The S value represents the sensor's sensitivity to external pressure and reflects the accuracy and effectiveness of the sensor. As shown in **Figure 7**(n), the sensor can reach a sensitivity of 0.24 kPa^{-1} even in a small stress range (<20 kPa). In Stage 1, the limited elastic deformation of the porous framework results in minimal magnetic–liquid–metal contact and a low rate of resistance change. With increasing pressure, pore collapse induces extensive magnetic liquid–metal contact and rapid conductive network densification, yielding a higher sensitivity (Stage 2). With further pressure loading, the conductive pathways approach saturation, leading to a decrease in the slope of resistance change (Stage 3). In the pressure range of 0–80 kPa, the pressure-sensing device can also respond quickly to the gradient rise pressure, showing good elasticity and recovery (**Figure 7**(o)). **Figure S29** shows the response and recovery time of the pressure-sensing device, and the resistance changes rapidly after the strain is applied and removed. The response time and recovery time are 0.76 s and 0.84 s, respectively, which indicates that the sensor has fast real-time performance in obtaining resistance change signals. To evaluate the durability of the sensor, a long-term cyclic compression test was performed with a pressure of 40 kPa for 500 cycles at a loading rate of 2 $mm \cdot min^{-1}$ (**Figure S30**). The sensor exhibited a stable and highly repeatable electrical response without obvious signal degradation, indicating excellent cyclic stability and mechanical reliability for practical applications. In summary, the piezoresistive pressure sensor prepared by this method has good stability and reliability. To demonstrate the practical applicability of this strategy, a 3D-printed resin gripper based on LM-20%Fe@resin was designed and fabricated. The tips of the gripper were magnetized to enable magnetic actuation. Effective gripping was successfully achieved by placing a NdFeB magnet underneath for attraction. As shown in **Video S8**, the gripper can perform gripping actions stably and repeatedly, and can repeatedly grasp a toy car weighing approximately 7 g without significant performance degradation.

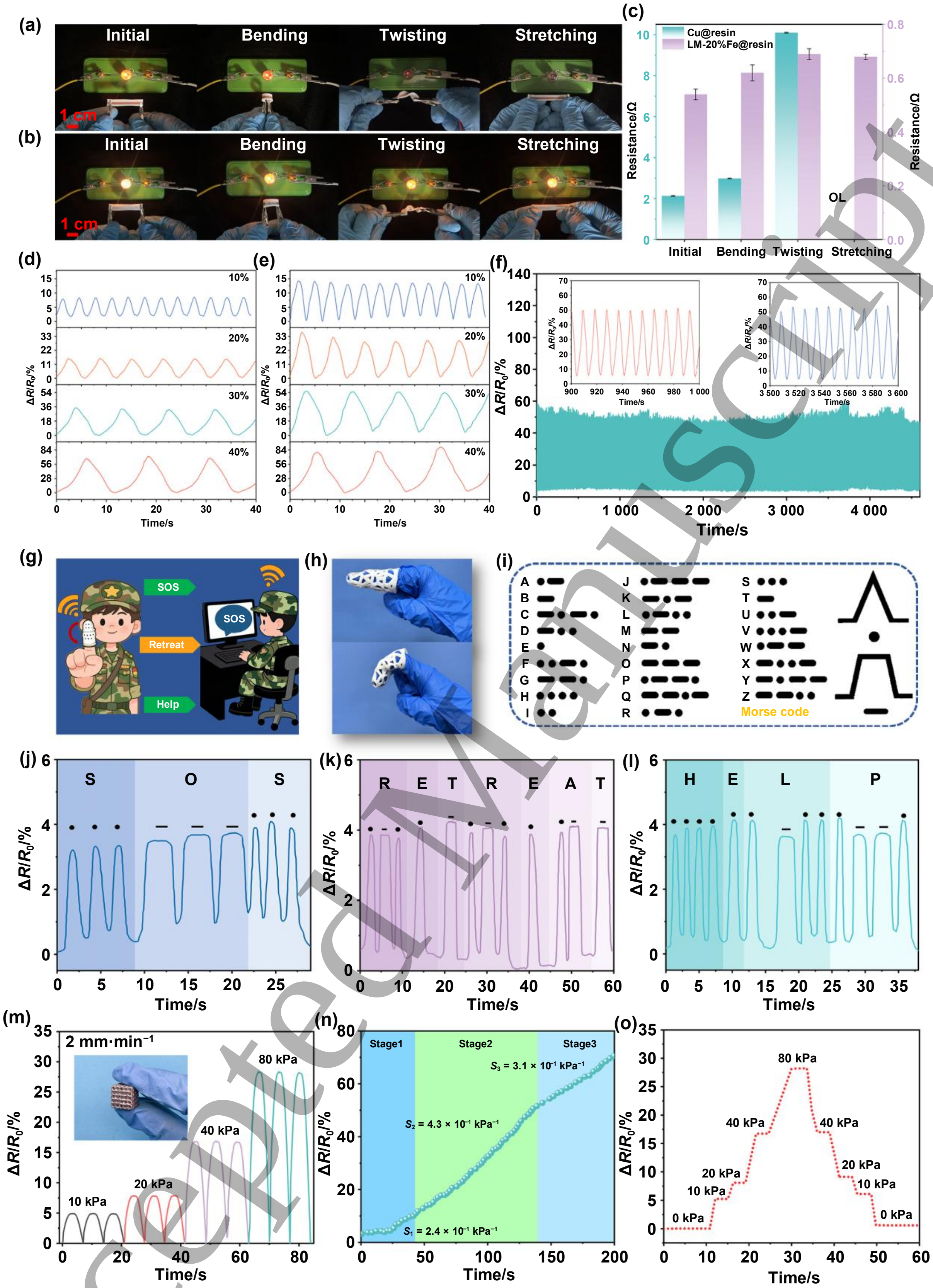

Figure 7. Photographs of the Cu@resin circuit **(a)** and LM-20%Fe@resin circuit **(b)** lighting up a small bulb under "initial," "bending," "twisting," and "stretching" states, with the corresponding resistance changes **(c)**. Relative resistance changes of the LM-0%Fe@resin **(d)** and LM-20%Fe@resin **(e)** strain sensors under 10%, 20%, 30%, and 40% strain. **(f)** Long-term stability of the LM-20%Fe@resin strain sensor for 500 cycles at 30% strain. **(g)** Conceptual diagram of spy communication using an LM-20%Fe@resin strain-sensing finger

cuff. **(h)** Photograph of the strain-sensing finger cuff. **(i)** The Morse code table. Relative resistance changes when "SOS" **(j)**, "RETREAT" **(k)**, and "HELP" **(l)** signals are output. **(m)** Relative resistance changes of a three-dimensional porous skeleton structure sensor based on LM-20%Fe@resin at different pressures. **(n)** The sensitivity of a three-dimensional porous skeleton structure sensor based on LM-20%Fe@resin under increasing pressure. **(o)** Relative resistance changes of a three-dimensional porous skeleton structure sensor based on LM-20%Fe@resin under pressure gradient changes.

3. Conclusions

In conclusion, we propose a strategy of LISM combined with magnetic liquid metal slurry infiltration, enabling selective high-precision magnetic patterning on various 3D-printed devices. The magnetic slurry is formed by mixing the magnetic material, Fe particles, into liquid metal, which serves as a deformable host for selective magnetic patterning. By alloying the liquid metal with the Cu layer, the patterned conductive metal is successfully imparted with magnetic properties. The obtained LM-20%Fe@resin has excellent light-thermal and magnetic-thermal conversion capabilities. On this basis, we successfully prepared 3D-printed small torpedoes that can be magnetically and photothermally driven through magnetizing the head or the tail for rapid and arbitrary route movement in water. In addition, by alternating the magnetic field strength, the directional movement of the 3D-printed caterpillar on the plane and the slope surface is realized. LM-20%Fe@resin has excellent antibacterial and EMI shielding capabilities, exhibits negligible cytotoxicity, and shows enhanced sensing sensitivity as a sensor due to the added Fe particles. Based on this, we have successfully prepared a wearable strain finger cuff for strain sensing to decrypt Morse codes, and a device with a three-dimensional porous skeleton structure for piezoresistive pressure sensing. The 3D printing enables user-specific customization, and the strategy of combining LISM and magnetic liquid metal slurry infiltration can achieve magnetic patterning on the surface of various printed devices, making it possible to personalize high-precision magnetic products.

4. Experimental methods

4.1. Materials

Photosensitive resin, Elastic FLGPCL04 based on acrylic acid, was purchased from Formlabs Ltd. The laser sensitizer nano zinc oxide (ZnO) was obtained from Shanghai Aladdin Biochemical Technology Co., Ltd. The analytical reagent grade copper sulfate ($CuSO_4$), potassium tartrate ($KNaC_4H_4O_6 \cdot 4H_2O$), methanol (CH_3OH), formaldehyde solution (HCHO), ethylenediaminetetraacetic acid (EDTA, $C_{10}H_{16}N_2O_8$), absolute ethanol

(CH_3CH_2OH) and hydrochloric acid (HCl) required for electroless copper plating were all provided by Chengdu Kelon Reagent. Liquid metal (Galinstan, melting point 11 °C) was purchased from Dongguan Dingguan Metal Technology Co., Ltd. Fe powder (particle size 20 nm) was purchased from Yongxin Alloy Co., Ltd. NdFeB permanent magnets were obtained from Ganzhou Permanent Magnet Electronics Co., Ltd. Commercially available conductive copper foil was obtained from Kunshan Shengshijing New Materials Co., Ltd. The artificial sweat was purchased from Hangzhou Fuwoke Biotechnology Co., Ltd.

4.2. 3D printing soft devices

Nano ZnO was added to the photosensitive resin Elastic FLGPCL04 and stirred with a high-speed stirrer for 2 h to obtain the 3D-printed resin that contained laser sensitizers (ZnO@resin). The ZnO@resin was printed using a commercial SLA 3D printer (Formlabs), where the 3D model was designed using Cinema 4D combined with Blender software and then imported into Preform software for printing size adjustment. The light-curing source of the 3D printer is a 405 nm UV laser with a spot size of 140 μm. The resin temperature was maintained at 35 °C, the thickness of the fixed layer was set at 25 μm, the energy density of the model filling exposure was 200.0 $mJ\cdot cm^{-2}$, and the waiting time after laser curing was 1 s. After printing, the uncured resin was washed off with anhydrous ethanol and then cured with UV light for 30 min at room temperature.

4.3. UV laser activation

The surface of the 3D-printed ZnO@resin device was laser activated using an MUV-E-A laser machine with a UV wavelength of 355 nm and a spot size of 20 μm. The arbitrary circuits (or patterns) to be processed were designed directly by EZCAD 2.0 software. In this experiment, the optimal scanning speed of the UV laser is 500 $mm\cdot s^{-1}$, which means that the laser beam is scanned through 500 mm in 1 s. The optimal laser parameters of the ZnO@resin were explored by changing the laser frequency and laser power. Finally, the laser-activated 3D-printed devices were ultrasonically cleaned in anhydrous ethanol for 5 min to remove the debris generated on the device surface during the laser process.

4.4. Electroless copper plating (ECP)

The laser-activated resins were reactively metallized in a configured ECP solution. The composition of the ECP solution was 8 $g\cdot L^{-1}$ $CuSO_4$, 30 $g\cdot L^{-1}$ potassium sodium tartrate, 3 $g\cdot L^{-1}$ EDTA, 3 $g\cdot L^{-1}$ $KNaC_4H_4O_6\cdot 4H_2O$, 12 $mL\cdot L^{-1}$ HCHO and 150 $mL\cdot L^{-1}$ CH_3OH. After preparation, NaOH was added to adjust the pH of the solution to 12.5. The laser-activated ZnO@resin was subjected to ECP at 49 °C for approximately

30 min. After the reaction, the 3D-printed ZnO@resin devices were washed with deionized water for 2 min to remove the excess ECP residue to obtain the Cu@resin devices.

4.5. Fabrication and selective infiltration of magnetic liquid metal slurries

Liquid metal magnetic slurries were obtained by adding 10 wt%, 20 wt%, and 30 wt% Fe powders to 50 mL of HCl solution (2 mol·L^{-1}) with constant stirring until the liquid metal almost completely encapsulated the Fe powders. The uncoated Fe powder on the surface of the liquid metal magnetic slurry surface was cleaned off using 0.5 mol·L^{-1} HCl solution, and then transferred to a Petri dish containing 20 mL of 0.5 mol·L^{-1} HCl solution. The area of the Cu@resin device containing the Cu pattern was immersed in the Petri dish, and a nylon brush was used to move the liquid metal magnetic slurry to the Cu area and keep brushing it. The liquid metal magnetic slurry selectively adhered to the Cu area on the Cu@resin device surfaces because of the alloying reaction between the liquid metal and the Cu layer in a short time. Afterward, the devices were continuously washed with deionized water to remove the residual HCl solution and then dried at room temperature to obtain complete magnetically patterned LM-Fe@resin devices.

4.6. Characterizations

The morphology of the nano ZnO particles was observed by a JEM-2100Plus field emission transmission electron microscopy (TEM) and the particle size was determined by Nano Measure software. The cured resin was washed after UV light irradiation, and the curing depth of the resin was measured using a high-precision thickness gauge (German). The magnetization curves of the different resin samples were determined using a LakeShore-7404 vibrating sample magnetometer (USA). The square resistance value was obtained by an HPS2524-coated square resistance tester (China). X-ray photoelectron spectroscopy (XPS) measurements were acquired by a multifunctional surface analysis electron spectrometer (Kratos, UK, XSAM800) equipped with an Al Kα X-ray source (1 486.6 eV). Raman spectroscopy was conducted on a Thermo Fisher Scientific DXRxi micro-Raman spectrometer (USA) with a 532 nm laser. Optical microscopy (OM) was used to observe the surface structure using a Keyence VHX-1000C digital microscope. Field-emission scanning electron microscopy (FE-SEM) images were acquired on an FEI Quanta-250 microscope (USA) at 10 kV equipped with an integrated energy-dispersive spectroscopy (EDS) detector for elemental analysis. X-ray diffraction (XRD) analysis was performed using a Rigaku D/max-2500 X-ray diffractometer (Japan) operated at 40 kV and 40 mA. Laser scanning confocal microscopy (LSCM) was conducted on a Zeiss LSM 710 system. The water contact angle of the samples before and after laser activation was measured by a Crewe DSA30 contact angle tester (German). Electromagnetic heating experiments were conducted by placing samples on a ZDBT-2 magnetic coil device (China) and setting different output powers. The samples were driven by irradiating them with an 808 nm near-infrared (NIR) laser, and characterizing the light-thermal effects of different samples, while a FLIR ONE PRO infrared camera was used to record the temperature change and images. The electromagnetic interference (EMI) shielding performance was evaluated using a vector network analyzer in the 8.0–12.5 GHz frequency range. The sensing performance of the resin was tested by a UT181A digital multimeter (China) with a Keithley 6487 voltage source. The antibacterial activity of the samples against *Staphylococcus aureus* (*S. aureus*) and *Escherichia coli* (*E. coli*) was evaluated by incubating 20 g of resin samples with bacterial suspensions (10^6 CFU·mL^{-1}) at 37 °C for 2 h. After 24 h of incubation, they were spread on solid media, and the total viable count (TVC) was determined. Cytotoxicity evaluations were tested using the CCK-8 assay and L929 cell morphology observation. Sterilized resin samples (1 mm × 1 mm, ^{60}Co-irradiated) were incubated in media for 24 h. L929 cell suspensions were mixed with 100 μL of culture for 1–5 days (37 °C, 5% CO_2), and cell suspensions without culture were used as controls. The cell viability was calculated using a microplate reader (Model 550, Bio-Rad Corp) at 450 nm. Live/dead staining was conducted using AM/PI dye, while cytoskeletal morphology was observed by fluorescein isothiocyanate-phalloidin labelling, both performed according to the manufacturer's protocols. Mechanical vibration testing was conducted using a small vibration test platform equipped with C-220 frequency-modulation control systems (China). Resistance testing at extreme temperatures was conducted using a freezer (−18 °C) and a high-temperature oven (90 °C).

Supporting information

The online version contains supplementary material available at .

Acknowledgments

The authors thank the National Natural Science Foundation of China (Grant Nos. 52573230, 52073194), Advanced Materials-National Science and Technology Major Project (Grant No. 2024ZD0607400), Sichuan Science and Technology Program (Grant Nos. 2026NSFSCZY0008, 2022JDRC0020), the Program for Featured Directions of Engineering Multidisciplines of Sichuan University (Grant No. 2020SCUNG203), and State Key Laboratory of Advanced Polymer Materials (Grant No. sklapm2025-1-11).

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflict of interest

The authors declare that they have no conflicts of interest.

References

[1] Huang X M, Peng S Q, Zheng L H, Zhuo D X, Wu L X and Weng Z X. 2023. 3D printing of high viscosity UV-curable resin for highly stretchable and resilient elastomer. *Adv. Mater*. **35**, 2304430.

[2] Sun X L, Chen S Y, Qu B, Wang R, Zheng Y Y, Liu X Y, Li W J, Gao J H, Chen Q H and Zhuo D X. 2023. Light-oriented 3D printing of liquid crystal/photocurable resins and in-situ enhancement of mechanical performance. *Nat. Commun*. **14**, 6586.

[3] Chen C, Yu H T, Lai T, Guo J, Qin M M, Qu Z G, Feng Y Y and Feng W. 2023. Flexible and elastic thermal regulator for multimode intelligent temperature control. *SusMat* **3**, 843-858.

[4] Wang C J, Sim K, Chen J, Kim H, Rao Z, Li Y H, Chen W Q, Song J Z, Verduzco R and Yu C J. 2018. Soft ultrathin electronics innervated adaptive fully soft robots. *Adv. Mater*. **30**, 1706695.

[5] Zhu G D, Hou Y, Xia N, Wang X, Zhang C, Zheng J Z, Jin D D and Zhang L. 2023. Fully recyclable, healable, soft, and stretchable dynamic polymers for magnetic soft robots. *Adv. Funct. Mater*. **33**, 2300888.

[6] Maurin V, Chang Y L, Ze Q, Leanza S, Wang J and Zhao R R. 2024. Liquid crystal elastomer–liquid metal composite: ultrafast, untethered, and programmable actuation by induction heating. *Adv. Mater*. **36**, 2302765.

[7] Wang M, Cheng Z W, Zuo B, Chen X M, Huang S and Yang H. 2020. Liquid crystal elastomer electric locomotives. *ACS Macro Lett*. **9**, 860-865.

[8] Yan Q *et al*. 2025. Precise engineering of asymmetric tri-active sites by symbiotic strategy for photocontrolled directional reforming of biomass. *Angew Chem. Int. Ed*. **64**, e202505718.

[9] Guo R, Sun X Y, Yuan B, Wang H Z and Liu J. 2019. Magnetic liquid metal (Fe-EGaIn) based multifunctional electronics for remote self-healing materials, degradable electronics, and thermal transfer printing. *Adv. Sci*. **6**, 1901478.

[10] Joyee E B and Pan Y Y. 2019. A fully three-dimensional printed inchworm-inspired soft robot with magnetic actuation. *Soft Robot* **6**, 333-345.

[11] Kim Y and Zhao X H. 2022. Magnetic soft materials and robots. *Chem. Rev*. **122**, 5317-5364.

[12] Zhang C Q, Li X J, Jiang L M, Tang D F, Xu H, Zhao P, Fu J Z, Zhou Q F and Chen Y. 2021. 3D printing of functional magnetic materials: from design to applications. *Adv. Funct. Mater*. **31**, 2102777.

[13] Kwok S W, Morin S A, Mosadegh B, So J H, Shepherd R F, Martinez R V, Smith B, Simeone F C, Stokes A A and Whitesides G M. 2014. Magnetic assembly of soft robots with hard components. *Adv. Funct. Mater*. **24**, 2180-2187.

[14] Ju Y W, Hu R, Xie Y, Yao J P, Li X X, Lv Y L, Han X T, Cao Q L and Li L. 2021. Reconfigurable magnetic soft robots with multimodal locomotion. *Nano Energy* **87**, 106169.

[15] Lee Y *et al*. 2023. Magnetically actuated fiber-based soft robots. *Adv. Mater*. **35**, 2301916.

[16] Wang H M, Zhu Z S, Jin H, Wei R, Bi L and Zhang W L. 2022. Magnetic soft robots: design, actuation, and function. *J. Alloy. Compd*. **922**, 166219.

[17] Wu S, Hamel C M, Ze Q, Yang F Y, Qi H J and Zhao R K. 2020. Evolutionary algorithm-guided voxel-encoding printing of functional hard-magnetic soft active materials. *Adv. Intell. Syst.* **2**, 2000060.

[18] Zheng H M, Rao D N, Sun H N, Li S S, Zhang C Q, Zhou F H, Mao L Y, Zhang H, Xie H and Li T F. 2025. Magnetic turtle-like robot with biomimetic movements through programmable magnetic-assisted 3D printing. *Small* **21**, 2412599.

[19] Xie Y, Zhang H Y, Zhang J H and Zhou T. 2022. Ultra-foldable integrated high-performance in-plane micro-supercapacitors from laser-induced selective metallization. *Energy Storage Mater.* **51**, 139-148.

[20] Xu H R, Luo P G, Wang J X, Li S Y and Zhou T. 2024. Laser-induced selective metallization directly preparing repairable superhydrophobic copper layer on polymers and single-droplet ethanol detection. *Chem. Eng. J.* **496**, 154228.

[21] Yu F F, Xu H R and Zhou T. 2024. Laser-induced selective metallization of epoxy resin: Preparing ultra-thin and ultra-light motors. *Compos. Part A: Appl. Sci. Manuf.* **177**, 107946.

[22] Zhang H Q, Wang P R, Chen Z, Chen X Y, Jiang M X, Yang J H and Li J. 2024. Functional metallic circuitries created by laser-activated selective electroless plating for 3D customized electronics. *Mater. Des.* **248**, 113513.

[23] Zhou Y, Meng Y N, Luo G H, Chen B, Zhong D P and Hu Y X. 2024. Laser-induced stress-driven nanoplate jumping visualized by ultrafast electron microscopy. *ACS Nano* **18**, 19783−19792.

[24] Cao L X, Yu D H, Xia Z S, Wan H Y, Liu C K, Yin T and He Z Z. 2020. Ferromagnetic liquid metal putty-like material with transformed shape and reconfigurable polarity. *Adv. Mater.* **32**, 2000827.

[25] Li P Y *et al*. 2025. Magnetointeractive Cr_2Te_3-coated liquid metal droplets for flexible memory arrays and wearable sensors. *Adv. Mater.* **37**, 2414519.

[26] Wu H Z, Luo R Y, Li Z F, Tian Y J, Yuan J Y, Su B, Zhou K, Yan C Z and Shi Y S. 2024. Additively manufactured flexible liquid metal–coated self-powered magnetoelectric sensors with high design freedom. *Adv. Mater.* **36**, 2307546.

[27] Xu Y, Zhu J Q, Chen H, Yong H C and Wu Z G. 2023. A soft reconfigurable circulator enabled by magnetic liquid metal droplet for multifunctional control of soft robots. *Adv. Sci.* **10**, 2300935.

[28] Xiong M F, Gao Y X and Liu J. 2014. Fabrication of magnetic nano liquid metal fluid through loading of Ni nanoparticles into gallium or its alloy. *J. Magn. Magn. Mater.* **354**, 279-283.

[29] Daalkhaijav U, Yirmibesoglu O D, Walker S and Mengüç Y. 2018. Rheological modification of liquid metal for additive manufacturing of stretchable electronics. *Adv. Mater. Technol.* **3**, 1700351.

[30] He X K, Ni M Y, Wu J P, Xuan S H and Gong X L. 2021. Hard-magnetic liquid metal droplets with excellent magnetic field dependent mobility and elasticity. *J. Mater. Sci. Technol.* **92**, 60-68.

[31] Li Y, Wu T, Wang Y J, Li J W, Yu W Q, Zhang G X, Chang B, Zhao L L, Liu H and Zhou W J. 2023. Green and large-scale production of ammonia: laser-driven pyrolysis of nitrogen-enriched biomass. *SusMat* **3**, 533-542.

[32] Cao Y J, Fan L L, Gao J Y, Zhu X Y, Wu B J, Wang H Z, Wang B, Shi J and Liu J. 2024. Magnetic and injectable Fe-doped liquid metals for controlled movement and photothermal/electromagnetic therapy. *J. Mater. Chem. B* **12**, 2313-2323.

[33] Chen H D *et al*. 2025. Thermomagnetic liquid metal switches with fast bidirectional response. *Nat. Commun.* **16**, 2634.

[34] Wang X L, Yao W H, Guo R, Yang X H, Tang J B, Zhang J, Gao W P, Timchenko V and Liu J. 2018. Soft and moldable Mg-doped liquid metal for conformable skin tumor photothermal therapy. *Adv. Healthc. Mater.* **7**, 1800318.

[35] Guo Z H *et al*. 2021. Galvanic replacement reaction for in situ fabrication of litchi-shaped heterogeneous liquid metal-Au nano-composite for radio-photothermal cancer therapy. *Bioact. Mater.* **6**, 602-612.

[36] Zhang W Q, Guo H J, Zhang X and Li H L. 2025. Stretchable and pH-sensitive liquid metal/mxene hydrogels for antibacterial environment creation and photothermal therapy of infected wounds. *Mater. Today Commun.* **47**, 112976.

[37] Saborio M G *et al*. 2020. Liquid metal droplet and graphene co-fillers for electrically conductive flexible composites. *Small* **16**, 1903753.

[38] Zhang J Z, Xu B G, Chen K L, Li Y, Li G and Liu Z K. 2024. Revolutionizing digital healthcare networks with wearable strain sensors using sustainable fibers. *SusMat* **4**, e207.

[39] Choe M, Sin D, Bhuyan P, Lee S, Jeon H and Park S. 2022. Ultrasoft and ultrastretchable wearable strain sensors with anisotropic conductivity enabled by liquid metal fillers. *Micromachines* **14**, 17.

[40] Deng H, Sattari K, Xie Y C, Liao P, Yan Z and Lin J. 2020. Laser reprogramming magnetic anisotropy in soft composites for reconfigurable 3D shaping. *Nat. Commun.* **11**, 6325.

[41] Zhong Y D, Tang W, Zhang C, Jiao Z D, Wu D M, Liu W T, Yang H Y and Zou J. 2022. Programmable thermochromic soft actuators with "two dimensional" bilayer architectures for soft robotics. *Nano Energy* **102**, 107741.

[42] Luo P G, Li S Y, Xu H R, Yu F F and Zhou T. 2025. Biomimetic superhydrophobic metal layers on 3D-printed resin via laser for flexible electronics and magnetic droplet manipulation. *Small* **21**, 2502756.

[43] Wu X *et al*. 2025. CeO_x-Integrated dual site enhanced urea electrosynthesis from nitrate and carbon dioxide. *Nat. Commun.* **16**, 8785.

[44] Xu H R, Yu F F, Feng J, Wang J X and Zhou T. 2023. Ultraviolet laser-induced selective metallization on white polymers using oxygen vacancy laser sensitizer for IoT sensors. *Ind. Eng. Chem. Res.* **62**, 20741-20751.

[45] Meng K, Zhang J J, Cheng B, Ren X G, Xia Z S, Xu F Y, Zhang L Y and Yu J G. 2024. Plasmonic near-infrared-response S-scheme $ZnO/CuInS_2$ photocatalyst for H_2O_2 production coupled with glycerin oxidation. *Adv. Mater.* **36**, 2406460.

[46] Xiao C C, Feng J, Xu H R, Xu R and Zhou T. 2022. Scalable strategy to directly prepare 2D and 3D liquid metal circuits based on laser-induced selective metallization. *ACS Appl. Mater. Interfaces* **14**, 20000-20013.

[47] Kim H, Lee K, Oh J W, Kim Y, Park J E, Jang J, Lee S W, Lee S, Koo C M and Park C. 2023. Shape-deformable and locomotive MXene ($Ti_3C_2T_x$)-encapsulated magnetic liquid metal for 3D-motion-adaptive synapses. *Adv. Funct. Mater.* **33**, 2210385.

[48] Kim M, Lim H and Ko S H. 2023. Liquid metal patterning and unique properties for next-generation soft electronics. *Adv. Sci.* **10**, 2205795.

[49] Yang J H, Cao J Q, Han J, Xiong Y, Luo L, Dan X Z, Yang Y J, Li L L, Sun J and Sun Q J. 2022. Stretchable multifunctional self-powered systems with Cu-EGaIn liquid metal electrodes. *Nano Energy* **101**, 107582.

[50] Wang Q, Li Q and Zhang J W. 2025. Effect of boron gradient design on the microstructure and properties of laser-clad Fe-based coatings. *Mater. Lett.* **398**, 138932.

[51] Zhou R H, Huang J Y, Li Z M, Wang Y S, Chen Z Y and Pan C F. 2024. High precision patternable liquid metal based conductor and adhesive substrate enabled stretchable hybrid systems. *Nano Res.* **17**, 5595-5603.

[52] Zhang H B *et al*. 2024. Light and magnetism orchestrating aquatic pollutant-degradation robots in programmable trajectories. *Adv. Mater.* **36**, 2311446.

[53] Pan M, Liu M M, Lei J Y, Wang Y Y, Linghu C H, Bowen C and Hsia K J. 2025. Bioinspired mechanisms and actuation of soft robotic crawlers. *Adv. Sci.* **12**, 2416764.

[54] Yi H, Jiang X Y, Feng L, Tian L F and Yang H M. 2024. High-frequency ultrasound modulation of Zn^{2+} release from nanoclay supported ZnO antibacterial composites. *Ultrason. Sonochem.* **111**, 107096.

[55] Elbourne A *et al*. 2020. Antibacterial liquid metals: biofilm treatment *via* magnetic activation. *ACS Nano* **14**, 802-817.

[56] Chen S, Huang F, Mao L J, Zhang Z M, Lin H, Yan Q X, Lu X Y and Shi J L. 2025. High Fe-loading single-atom catalyst boosts ROS production by density effect for efficient antibacterial therapy. *Nano-Micro Lett.* **17**, 32.

[57] Wang L Y, Lai R Z, Zhang L C, Zeng M Q and Fu L. 2022. Emerging liquid metal biomaterials: from design to application. *Adv. Mater.* **34**, 2201956.

[58] Zhang X L, Deng Z S, Song H Z, Guo M H and Li L. 2024. Liquid metal electromagnetic wave shielding and absorbing film for solving electromagnetic interference in flexible sensors. *Sci. China Mater.* **67**, 3976-3985.

[59] Kim M G, Lee B, Li M C, Noda S, Kim C, Kim J, Song W J, Lee S W and Brand O. 2020. All-soft supercapacitors based on liquid metal electrodes with integrated functionalized carbon nanotubes. *ACS Nano* **14**, 5659-5667.